MILITARY SPECIFICATION SHEET

PLASTIC SHEET, LAMINATED, METAL-CLAD (FOR PRINTED WIRING BOARDS), BASE MATERIAL GR, GLASS BASE, NONWOVEN, POLYTETRAFLUOROETHYLENE RESIN, FLAME RESISTANT, FOR MICROWAVE APPLICATION, COPPER-CLAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the plastic sheets described herein shall consist of this specification and the latest issue of MIL-P-13949.

REQUIREMENTS:

Property values: See table I.

TABLE I. Property values.

Property to be tested	Test condition 1/	Thickness (inch)			
		Under 0.030 copper types 5/		0.030 and over copper types 5/	
		A B E F	C D	A B E F	C D
1. Peel strength, minimum in lbs-per-inch width					
A. Under 1/2-oz/ft^2 copper					
As received	A	---	5.0	---	6.0
After thermal stress	A	---	5.0	---	6.0
At elevated temperature	150°C	---	4.0	---	5.0
After exposure to processing solutions	A	---	4.0	---	6.0
B. 1/2-oz/ft^2 copper					
As received	A	4.0	5.0	4.0	6.0
After thermal stress	A	4.0	5.0	4.0	6.0
At elevated temperature	150°C	3.0	4.0	3.0	5.0
After exposure to processing solutions	A	3.0	4.0	3.0	5.0
C. 3/4-oz/ft^2 copper					
As received	A	4.0	6.0	4.0	7.0
After thermal stress	A	4.0	6.0	4.0	7.0
At elevated temperature	150°C	3.0	5.0	3.0	6.0
After exposure to processing solutions	A	3.0	5.0	3.0	6.0
D. 1-oz/ft^2 copper					
As received	A	5.0	7.0	5.0	8.0
After thermal stress	A	5.0	7.0	5.0	8.0
At elevated temperature	150°C	4.0	5.0	4.0	6.0
After exposure to processing solutions	A	4.0	5.0	4.0	6.0

See footnotes at end of table.

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TABLE I. Property values - Continued.

Property to be tested	Test condition 1/	Thickness (inch)			
		Under 0.030 copper types 5/		0.030 and over copper types 5/	
		A B E F	C D	A B E F	C D
E. 2-oz/ft^2 copper					
As received	A	6.0	8.0	7.0	10.0
After thermal stress	A	6.0	8.0	7.0	10.0
At elevated temperature	150°C	4.0	6.0	5.0	7.0
After exposure to processing solutions	A	4.0	6.0	5.0	7.0
F. 3-,4-, and 5-oz/ft^2 copper					
As received	A	---	---	9.0	12.0
After thermal stress	A	---	---	9.0	12.0
At elevated temperature	150°C	---	---	7.0	12.0
After exposure to processing solutions	A	---	---	7.0	12.0

Property to be tested	Test condition 1/	Thickness	Values required
2. Volume resistivity, minimum in megohm-centimeter			
After moisture resistance	F	All	10^6
At elevated temperature	E-24/150	All	10^5
3. Surface resistivity, minimum megohms			
After moisture resistance	F	All	10^4
At elevated temperature	E-24/150	All	10^3
4. Water absorption, maximum percent	E-1/105 +des +D-24/23	(inch)	
		0.031	0.10
		0.062	0.10
		0.093	0.10
		0.125	0.10
		0.250	0.10
5. Dielectric breakdown (parallel to laminations), minimum kV, step-by-step	D-48/50+ D-0.5/23	Applicable only for 0.031 inch and over.	30
6. Permittivity at X band 2/ 3/	C-24/23/50	All 6/	2.20 ±0.04 4/
		All	2.25 ±0.04
		All	2.30 ±0.04
		All	2.35 ±0.04
		All	2.40 ±0.04
7. Loss tangent at X band 2/	C-24/23/50	All 6/	0.0015 max
8. Q (resonance), minimum	N/A	N/A	N/A

See footnotes at end of table.

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TABLE I. Property values - Continued.

Property to be tested	Test condition 1/	Thickness	Values required
9. Flexural strength, minimum thousand lbs-per-square inch			
A. Lengthwise	A	Applicable only for 0.031 inch and over.	3.5
B. Crosswise	A		3.5
C. Lengthwise at elevated temperature	N/A	N/A	N/A
10. Arc resistance, minimum seconds	D-48/50+ D-0.5/23	120	180
11. Flammability, maximum burntime, 7/ seconds, each flame application	N/A	N/A	N/A

1/ See specimen conditioning requirements, conditioning tolerances and testing information table XXVI of MIL-P-13949.
2/ For the applicable test method see appendix A.
3/ Value shall be specified in ordering data.
4/ Tolerance of ±0.02 for all values of dielective constant may be specified in the ordering data.
5/ Copper types (see 1.2.1.1.3).
6/ Materials other than 0.030 and 0.060 inch thick are not testable at 10 GHz.
7/ PTFE materials will not burn in small scale flame test.

Thickness: Cut sheets less than 18 by 16 inches shall meet the applicable thickness tolerances in 100 percent of the area of the sheet.

Marking: Internal marking not applicable.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
NSA - S5
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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